

Project Documentation

SafeLA Analytics (2020–May 2025)

1. Project Planning & Management

1.1 Project Proposal

Project Title: Crime Analysis & Forecasting Dashboard (2020–May 2025)

1.2 Project Summary

This project analyzes crime incidents reported in Los Angeles from **January 2020 to May 2025**.

We cleaned, transformed, modeled, and visualized the data using multiple tools: Tableau, Power BI, Excel, Python, and forecasting models (CatBoost).

The final output is a set of interactive dashboards that help government agencies, police departments, and analysts identify crime trends, hotspots, victim demographics, weapon usage, and future crime forecasts.

1.3 Objective:

- Analyze crime patterns across time, location, and type.
 - Identify the most affected age groups and genders.
 - Understand weapon usage and arrest patterns.
 - Predict crime levels for the next 2 years.
 - Create interactive dashboards supporting decision-making.
-

1.4 Scope:

Included:

- Data cleaning
- Categorization of 140+ crime types
- Categorization of 300+ weapon types

- Exploratory analysis
- Dashboards in Tableau, Power BI, Excel
- Forecasting using CatBoost

Excluded:

- Real-time prediction
- Detailed case-level qualitative analysis

2. Team Roles & Responsibilities

Team Member	Task
Sara	Team Leader, Tableau dashboards, documentation, PPT, insights
Nariman	Excel dashboard, Data collection, Power Query cleaning, data dictionary
Moataz	Power BI dashboard, Python cleaning, notebooks, QA validation
Ziad	Python dashboard, EDA & insights generation, chart prototyping
Alaa	2-year forecasting model using CatBoost + Tableau forecast dashboard

3. Data Description

3.1 Dataset Source

Los Angeles Crime Dataset (2020–May 2025)

3.2 Data Dictionary

Column	Meaning / What It Represents
Incident_ID	Unique identifier for each crime report
Report_Date	Date crime was officially reported
Incident_Date	Actual date when the crime occurred
Incident_Time	Crime occurrence time
Area_Name	Name of the geographic reporting area
Crime_Type	Original detailed crime description (140+ values)
Victim_Age	Age of the victim
Victim_Gender	Gender of the victim
Premise_Type	Location type where crime occurred
Case_Status	Whether arrest happened or not
Weapon_Cleaned	Weapon category (after grouping 300+ weapons → 10 clean categories)
Latitude	GPS coordinate
Longitude	GPS coordinate

Year	Extracted crime year
Month	Extracted crime month
Day of Week	Day name (Monday–Sunday)
Report_Delay_Days	Days between incident & report
Time_Period	Categorized time (Morning/Afternoon/Evening/Night)
Victim_Age_Group	Age group category
Arrest_Flag	1 = Arrested, 0 = Not Detained
Month_Year	Monthly time index used for trends

4. Data Cleaning & Transformation

4.1 Cleaning Steps:

Step 1: Load Data

Load CSV file and verify it exists.

```
df = pd.read_csv('LADData.csv')
```

Step 2: Column Mapping

Standardize column names for consistency.

Mapping Examples:

- DR_NO → dr_no
 - Date Rptd → date_rptd
 - DATE OCC → date_occ
 - Vict Age → vict_age
-

Step 3: Column Selection

Keep 19 essential columns, drop 8 unnecessary ones.

Essential Columns (19):

Category	Columns
IDs & Location	dr_no, area, area_name, rpt_dist_no, location, cross_street, lat, lon
Date & Time	date_rptd, date_occ, time_occ
Crime Info	crm_cd, crm_cd_desc, status, status_desc
Victim Info	vict_age, vict_sex, vict_descent
Details	premis_desc, weapon_desc

Step 4: Missing Values Analysis

Document missing data patterns without deleting rows.

Common Patterns:

- weapon_desc: High missing (not all crimes involve weapons)
- cross_street: Moderate missing (not all locations at intersections)
- vict_age/sex/descent: Some missing (unknown victim info)

```
=====
4. MISSING VALUES ANALYSIS
=====
```

	Missing_Count	Percentage
crm_cd_4	1004927	99.993632
crm_cd_3	1002677	99.769749
crm_cd_2	935831	93.118346
cross_street	850755	84.652997
weapon_cd	677744	67.437818
weapon_desc	677744	67.437818
mocodes	151619	15.086603
vict_descent	144656	14.393761
vict_sex	144644	14.392567
premis_desc	588	0.058508
premis_cd	16	0.001592
crm_cd_1	11	0.001095
status	1	0.000100

Step 5: Date/Time Cleaning

5.1 Date Fields

Fields: date_rptd, date_occ

Invalid Criteria:

- X Future dates (> today)
- X Before 1900
- X Null/missin

5.2 Time Field

Field: time_occ (Military format: HHMM)

Validation:

- Must be 4 digits
- Hour: 00-23
- Minute: 00-59

5.3 Combined DateTime

Create datetime_occ by combining valid date + time for time-series analysis.

Step 6: Victim Age Cleaning

Valid Range: 0 to 120 years

Invalid if:

- Age < 0 (negative)
- Age > 120 (unrealistic)
- Null/missing

```
=====
6. CLEANING VICTIM AGE
=====
Age statistics:
  Min: -4
  Max: 120
  Mean: 28.92
  Median: 30.0
Invalid ages (< 0 or > 120 or null): 137

Top problematic age values:
vict_age
-1      100
-2       28
-3        6
-4         3
Name: count, dtype: int64
```

Step 7: Categorical Fields Standardization

7.1 Victim Sex

Valid Codes: F, M, X, NAN, NONE

Process:

1. Convert to uppercase
2. Trim whitespace
3. Validate against allowed values

```
=====
7. CLEANING CATEGORICAL FIELDS
=====
Vict Sex - Invalid values: 115
  Unique invalid values: ['H' '-']
  Value counts:
vict_sex
M      403879
F      358580
NAN    144644
X       97773
H        114
-         1
Name: count, dtype: int64
```

7.2 Victim Descent

Valid Codes: Single letters A-Z plus special codes

Common Codes:

- H = Hispanic/Latin/Mexican
- W = White
- B = Black
- A = Other Asian
- X = Unknown

- *[Full list: A,B,C,D,F,G,H,I,J,K,L,O,P,S,U,V,W,X,Z]*

```
Vict Descent - Invalid values: 0
```

```
Value counts:
```

```
vict_descent
```

```
H      296404
```

```
W      201442
```

```
NAN    144656
```

```
B      135816
```

```
X      106685
```

```
O       78005
```

```
A       21340
```

```
K        5990
```

```
F        4838
```

```
C         4631
```

```
J         1586
```

```
V         1195
```

```
I         1015
```

```
Z          577
```

```
P          288
```

```
U          221
```

```
D           91
```

```
L           77
```

```
G           74
```

```
S           58
```

```
-            2
```

```
Name: count, dtype: int64
```


Step 8: Geographic Data Cleaning

LA County Boundaries:

- Latitude: 33.7 to 34.35
- Longitude: -118.7 to -118.1

```
=====
8. CLEANING GEOGRAPHIC DATA
=====
Latitude range: 0.0000 to 34.3343
Longitude range: -118.6676 to 0.0000
Invalid coordinates: 2240
  Zero coordinates (0,0): 2240
  Out of LA bounds: 0
```

Step 9: Duplicate Detection

Check for duplicate DR_NO values (should be unique).

Action: Report count but don't delete (may be legitimate case updates)

Step 10: Crime Code Validation

Validate that crm_cd matches crm_cd_1.

Report:

- Count of mismatches
- Top 10 most common crime codes
- Crime type distribution

```
=====
10. VALIDATING CRIME CODES
=====
Crime code mismatches (crm_cd vs crm_cd_1): 1004991

Top 10 Crime Codes:
crm_cd
510    115190
624    74839
330    63517
354    62537
740    61092
310    57871
440    53717
230    53525
626    46712
420    41314
Name: count, dtype: int64
```

Step 11: Text Field Cleaning

Fields Cleaned: location, cross_street, area_name, premis_desc, weapon_desc, status_desc, crm_cd_desc

Operations:

1. Trim leading/trailing spaces
 2. Replace 'nan' strings with actual NaN
 3. Replace 'None' strings with NaN
-

Step 12: Area Validation

Valid Range: LAPD areas 1-21

Invalid if:

- Area < 1
- Area > 21
- Null/missing

The 21 Areas: Central, Rampart, Southwest, Hollenbeck, Harbor, Hollywood, Wilshire, West LA, Van Nuys, West Valley, Northeast, 77th Street, Newton, Pacific, N Hollywood, Foothill, Devonshire, Southeast, Mission, Olympic, Topanga

12. VALIDATING AREA CODES

Invalid area codes (should be 1-21): 0

Area distribution:

area

1	69670
2	46825
3	57441
4	37085
5	41394
6	52429
7	48239
8	45729
9	42883
10	42156
11	42963
12	61758
13	49177
14	59514
15	51107
16	33133
17	41756
18	49936
19	40351
20	50071
21	41374

Name: count, dtype: int64

Saved Files:

```
=====
15. SAVING CLEANED DATA
  Columns: 20 (essential columns only)
  Purpose: Production-ready for dashboards and analysis

✓ Saved: LADData_issues.csv
  Records: 2,492 (0.2% of total)
  Columns: 27 (20 data + 7 flags)
  Purpose: Manual review and data quality investigation

=====
FINAL SUMMARY
=====
Original records: 1,004,991
High quality records: 1,002,499 (99.8%)
Records with issues: 2,492 (0.2%)

Output files created (3 files):
  1. LADData_with_flags.csv
    - All records with quality flags
    - 27 columns
    - Use for: Quality control

  2. LADData_high_quality.csv ★ USE THIS
    - Clean records only (1,002,499 records)
    - 20 columns
    - Use for: Dashboards, visualizations, analysis

  3. LADData_issues.csv
    - Problematic records only (2,492 records)
    - 27 columns (includes flags)
    - Use for: Data quality review

✓ Data cleaning pipeline complete!
=====
```

Step 13: Handling Invalid Age Values (0)

Problem

Some rows contained an **Age = 0**, which is not a valid age.

Method

1. Used Power Query to calculate the **median age**, which was **30**.
2. Replaced all age values equal to **0** with the median (30).

Reasoning

- Median is more resistant to outliers than the mean.
- Ensures age distribution stays realistic.

Step 14: Fixing Missing Values in the Descent Column

Problem

The *Descent* column contained missing values (NaN).

Method

1. Identified the **mode** (most frequent value) using a **Pivot Table** in Excel.
2. Replaced all NaN values in *Descent* with that mode.

Reasoning

- Replacing categorical nulls with the mode preserves the distribution of categories.
 - Prevents category loss and reduces bias.
-

Step 15: Handling Null Values in the Sex Column

Problem

The *Sex* column contained null values.

Method

1. Used an Excel Pivot Table to count the distribution of Male vs. Female.
2. Calculated the most common value (mode), which was **Male**.
3. Replaced all null values in the *Sex* column with “**Male**”.

Reasoning

- For categorical binary variables, mode is the most appropriate imputation.
 - Prevents imbalance and keeps dataset consistent.
-

Step 16: Validation

All replacement steps were checked using:

- Preview of results in Power Query
 - Pivot Tables in Excel
 - Screenshots (recorded for documentation)
-

Step 17: Time Column Cleaning

The `time_occ` column contained inconsistent and invalid time formats such as missing digits, mixed characters, and impossible time values. To standardize this column, I wrote a Python function that automatically cleans and converts all entries into a uniform **HH:MM** format.

Steps Performed

- 1. Extract Only Numeric Characters**
 - a. Some values included symbols or text.
 - b. The function keeps only digits and removes everything else.
 - 2. Fix Incomplete Time Values**
 - a. Short values like 30 or 845 were padded with zeros to a 4-digit format.
 - i. Example: 30 → 0030, 845 → 0845.
 - 3. Split Into Hours and Minutes**
 - a. The first two digits become the hour, the last two the minutes.
 - 4. Validate Time**
 - a. Hours greater than 23 or minutes greater than 59 were marked as invalid and replaced with null.
 - 5. Convert to Standard Time Format (HH:MM)**
 - a. All valid values were formatted correctly (e.g., 0845 → 08:45).
 - 6. Convert Column to Datetime Type**
 - a. Ensures the column is sortable and ready for time-based analysis.
 - 7. Save the Cleaned Dataset**
 - a. The cleaned dataset was exported to a new CSV file.
-

4.2 Validation Rules

Summary Table

Field	Valid Criteria	Invalid Examples	Flag
Dates	1900 < date < today	Future dates, pre-1900, null	date_flag
Time	0000-2359 format	2400, 1265, abcd	time_valid
Age	0-120 years	-1, 150, null	age_flag
Sex	F, M, X, NAN, NONE	male, f, invalid	sex_flag
Descent	A-Z single letters	AA, invalid, null	descent_flag
Coordinates	LA County bounds	(0,0), out of range, null	coord_flag
Area	1-21	0, 25, null	<i>Reported only</i>

4.3 Output Files

File 1: LAData_with_flags.csv

Purpose: Quality control and debugging

Contents:

- All records (100%)
 - 26 columns: 19 data + 7 quality flags
 - Both clean and problematic records
-

File 2: LAData_high_quality.csv

Purpose: Production-ready for analysis and visualization

Contents:

- Clean records only (~90%)
 - 19 data columns (NO quality flags)
 - All validation checks passed
-

File 3: LAData_issues.csv

Purpose: Manual review and quality investigation

Contents:

- Problematic records only (~10%)
 - 26 columns: 19 data + 7 quality flags
 - Records that failed validation
-

4.4 Process Summary

Final Statistics

- **Input:** 27+ columns
 - **Output:** 19 data columns + 7 quality flags
 - **Clean Records:** ~90%
 - **Flagged Records:** ~10%
 - **Processing:** 14 steps, fully automated
-

4.5 Crime Type Grouping

We grouped **140+ messy crime descriptions** into **12 clean categories**:

New Category	Contains (Examples)
Assault / Violence	ASSAULT, BATTERY, THREATS, STALKING, KIDNAPPING
Theft / Larceny	THEFT, BURGLARY, SHOPLIFTING, PURSE SNATCHING
Robbery	ROBBERY, ATTEMPTED ROBBERY
Fraud / Forgery	FRAUD, FORGERY, EMBEZZLEMENT, COUNTERFEIT
Drugs / Narcotics	DRUGS, NARCOTICS
Weapons / Firearms	FIREARM, GUN, SHOTS FIRED, BOMB
Sex Crimes	RAPE, SODOMY, LEWD ACTS, PORNOGRAPHY
Child / Family Crimes	CHILD ABUSE, NEGLECT
Property Damage	VANDALISM, ARSON
Vehicle Crimes	VEHICLE THEFT, BIKE STOLEN
Public Disorder	RIOT, DISTURBANCE
Other / Unknown	The rest

4.6 Weapon Type Grouping

300+ weapons grouped into **10 categories**:

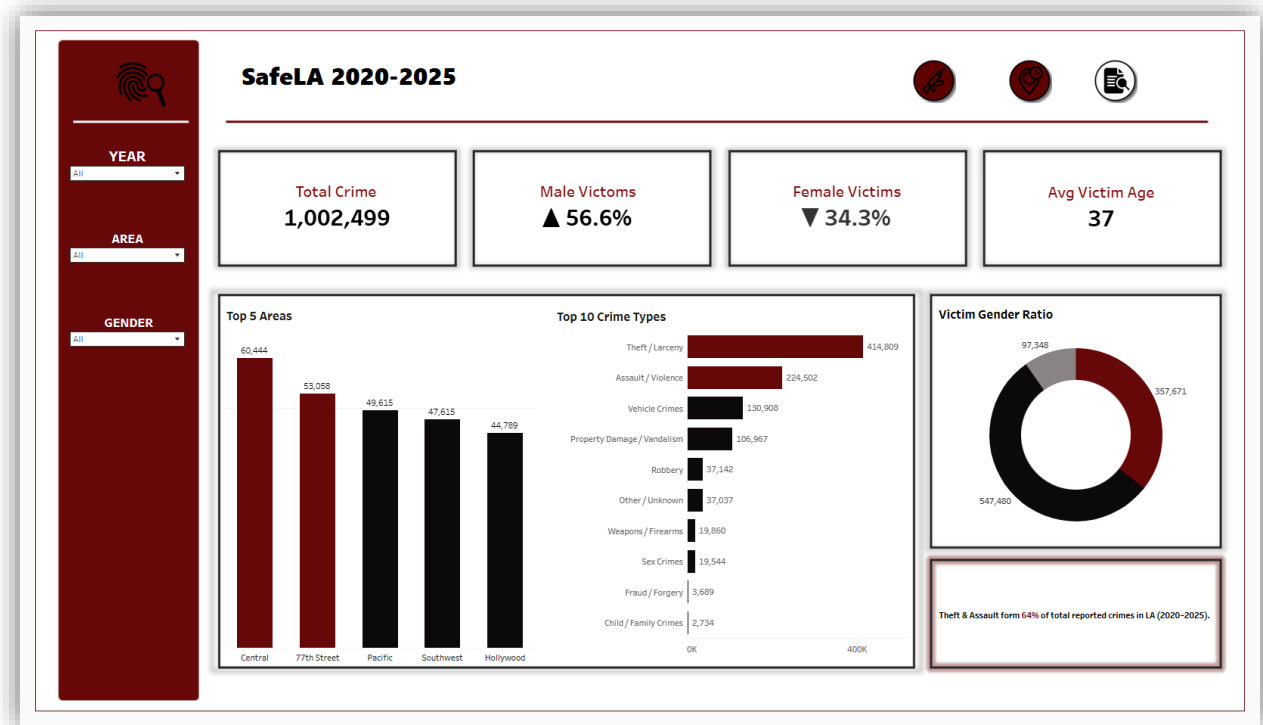
Category	Includes (Examples)
Firearm	Pistol, Rifle, Shotgun, Revolver
Sharp Object	Knife, Blade, Scissors
Heavy Object	Hammer, Brick, Pipe
Chemical	Acid, Caustic substances
Explosive	Bomb, Device
Strong-Arm	Hands, Kicks, Fists
Toy / Simulated	Fake gun
Vehicle	Car used as weapon
Animal	Dog
Unknown	Not reported

5. Dashboards Explanation

5.1 Tableau Story (Sarah)

❖ Page 1 — Crime Overview

Objective: Show the big picture.

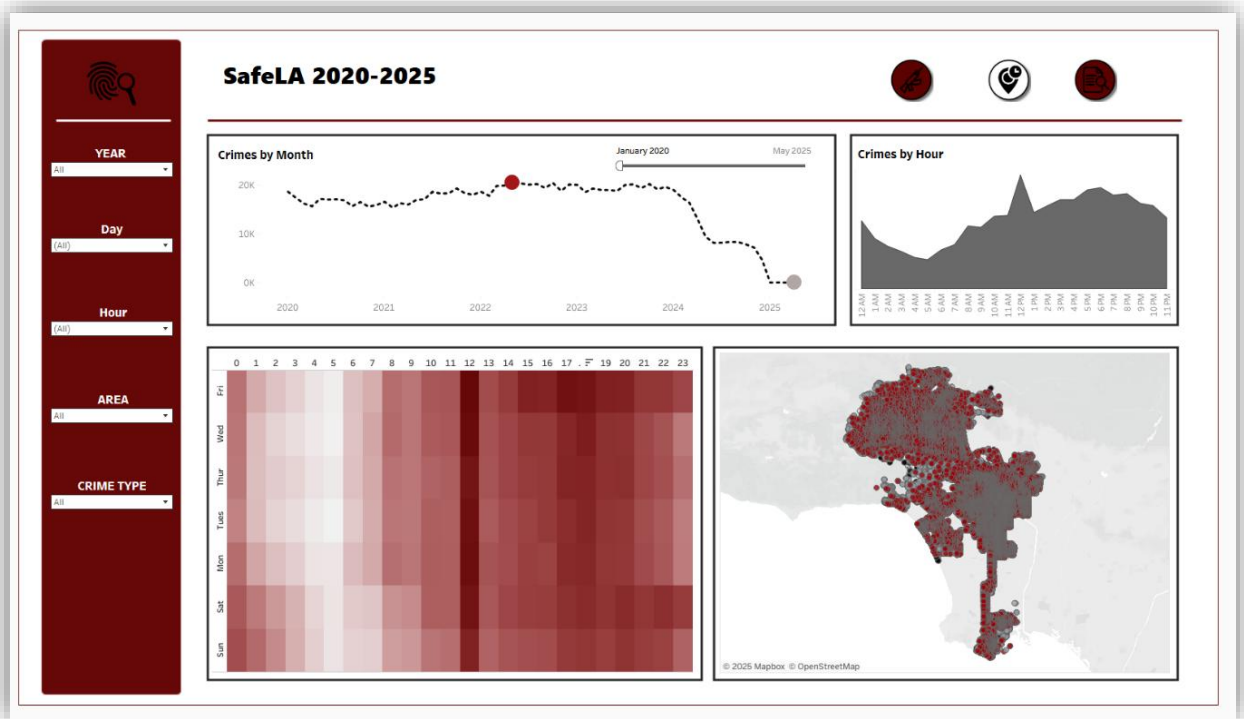


Charts:

- **KPI Cards** → Total Crimes, Avg Victim Age, %Male, %Arrested
- **Bar Chart** → Top 10 Crime Types
- **Bar Chart** → Top 10 Areas
- **Pie Chart** → Victim Gender
- **Insight Box** → “Assault + Theft represent 34% of total crimes (2020–May 2025).”

❖ Page 2 — Time & Location

Objective: When & where crimes happen.

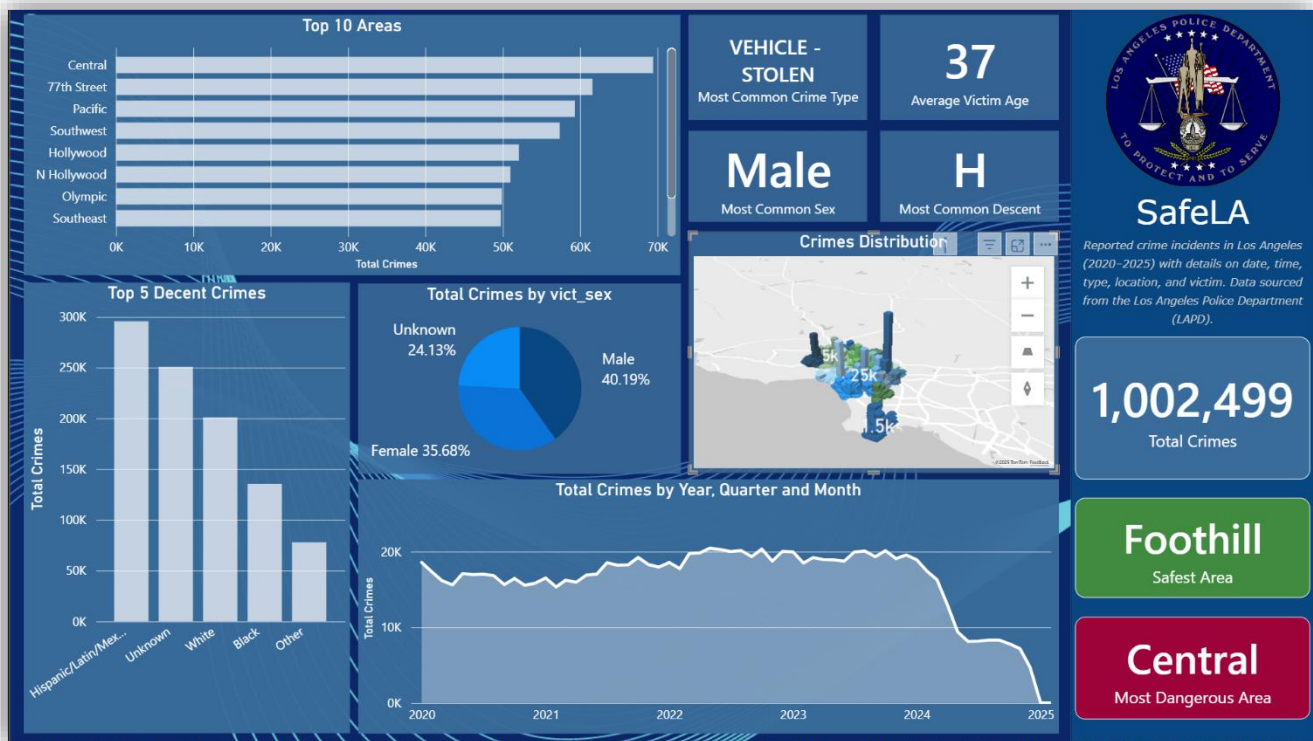


Charts:

- **Line Chart** → Monthly trends (crimes by month)
- **Histogram** → Crimes by hour
- **Heatmap** → Crimes by hour × weekday
- **Map** → Hotspot areas.

Objective: Who is affected & what happens after the crime.

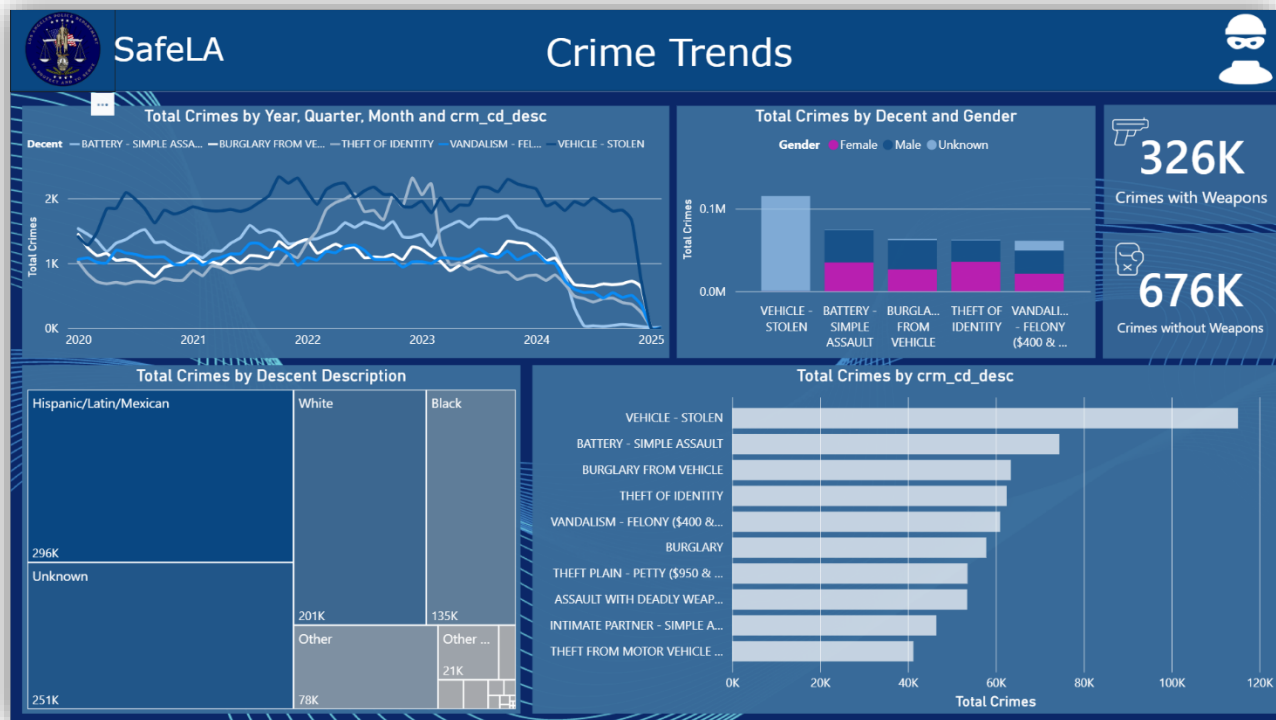
5.2 Power BI Dashboard (Moataz)



A. Overview Page

This page provides a general summary of crimes across the city.

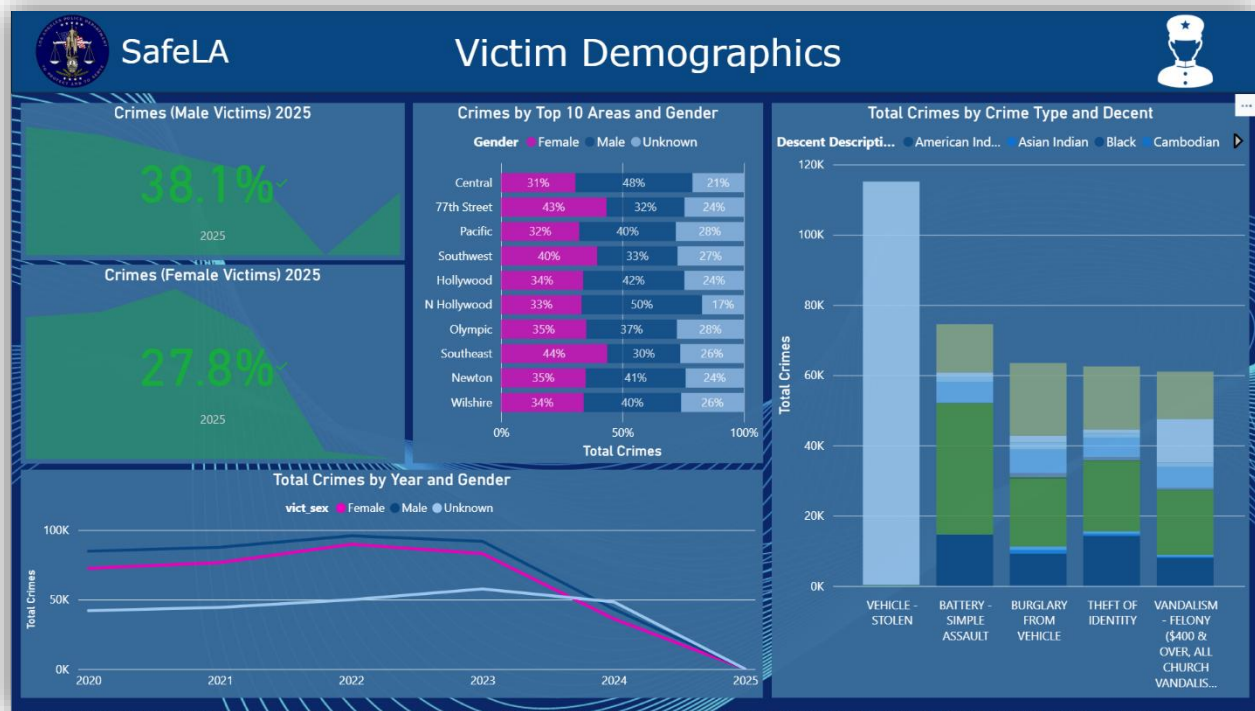
- **Top 10 Areas:** Ranks areas by total crime volume.
- **Most Common Crime Type:** Highlights the highest-occurring crime (e.g., Vehicle – Stolen).
- **Victim Insights:** Shows average victim age, most common sex, and most common descent.
- **Geographical Crime Distribution Map:** Interactive map displaying crime density.
- **Total Crimes Over Time:** Line chart showing yearly and monthly crime progression.
- **Safest & Most Dangerous Areas:** Automatically identifies the lowest and highest crime regions.



B. Crime Trends Page

Focuses on how crimes evolve over time and differences across groups.

- **Crimes by Type Over Time:** Line chart comparing trends in major crime categories.
- **Crimes by Descent & Gender:** Stacked bar chart summarizing victim characteristics.
- **Weapons Involvement Summary:** Total crimes involving weapons vs. without.
- **Descent Description Tree Map:** Shows the distribution of crimes by descent.
- **Top Crime Types:** Bar chart ranking the most frequent crime categories.



C. Victim Demographics Page

Provides demographic patterns among victims.

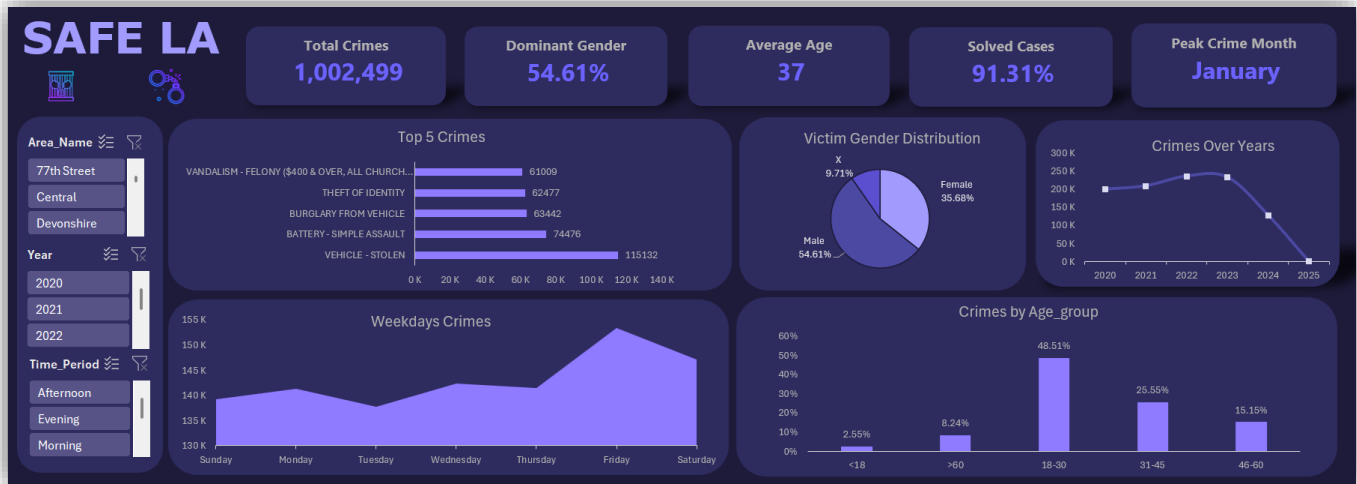
- **Male and Female Crimes (2025):** Summary cards showing percentages by gender.
- **Crimes by Area and Gender:** Horizontal stacked bars comparing gender distribution in the top crime areas.
- **Crimes by Crime Type & Descent:** Stacked bars showing demographic variations across different crime types.
- **Crimes by Year and Gender:** Line chart illustrating changes in victim gender distribution across the years.

Link to dashboard:

https://drive.google.com/drive/folders/1leXSpX6J672uMhnb6nOkxgA-mj0p-70p?usp=drive_link

5.3 Excel Dashboard (Nariman)

- Pivot tables
- Crime KPIs
- Year-wise summaries
- Gender distribution
- Area-level stats



A. Key Performance Indicators

- Total Crimes: 1,002,499
- Dominant Gender: Male (54.61%)
- Average Age: 37
- Solved Cases: 91.31%
- Peak Crime Month: January

B. Filters

- Area Name
- Year (2020–2022)
- Time Period (Morning, Afternoon, Evening)

C. Visual Insights

- **Top 5 Crimes:** Vehicle theft, simple assault, vehicle burglary, identity theft, felony vandalism
- **Gender Breakdown:** Male (54.61%), Female (35.68%), Other (9.71%)

- **Yearly Trend:** Crime increases until 2023–2024, then drops sharply in 2025, and that’s because 2025 data is not completed yet.
- **Weekday Trend:** Friday has the highest crime levels
- **Age Groups:** Most crimes involve individuals aged 18–30

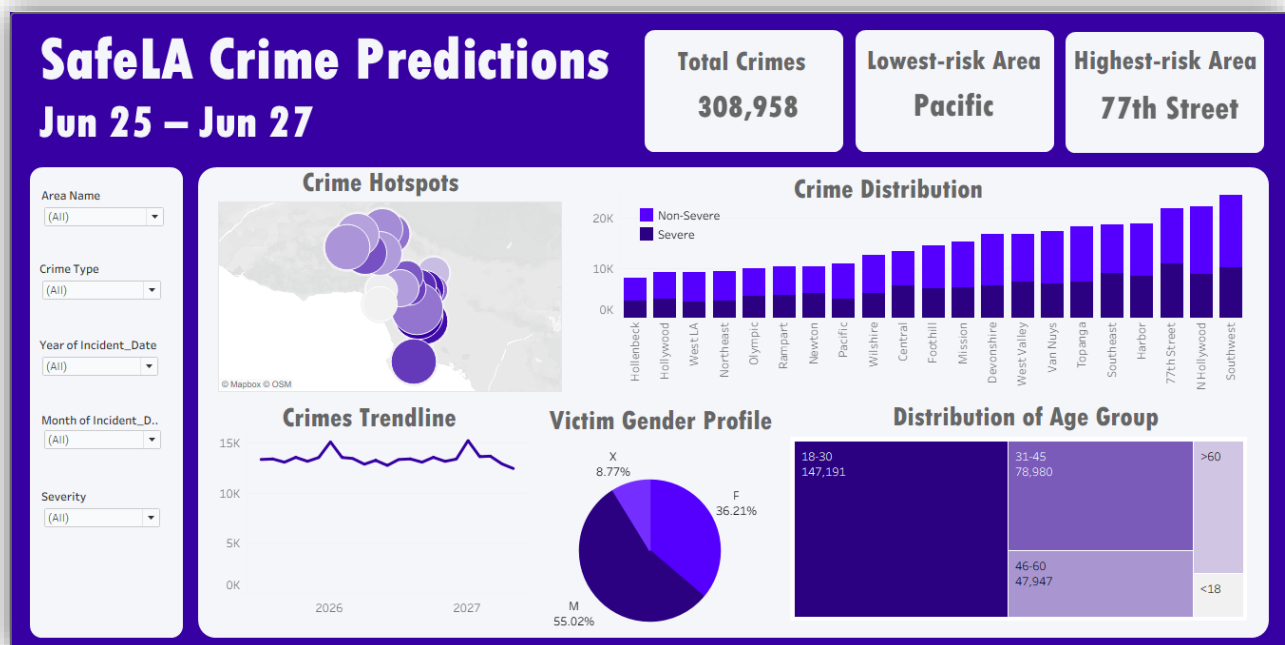
D. Key Insights

- Vehicle-related crime is the most prevalent category.
- Crime is most common among adults aged 18–30.
- Fridays consistently have the highest crime activity.
- January is the month with the highest number of crimes.

Link to dashboard:

https://drive.google.com/drive/folders/1k2eO6GKv46lG7hGOEo2DGRX8kOjmrCS?usp=drive_link

5.4 Forecasting Model (Alaa)



- **Preprocessing**
 - Grouped incidents by Area_Name × Crime_Type × Incident_Date.
 - Engineered time-series features: Year, Month, ISO Week, Day.
 - Added lag features (lag_1, lag_2) for each area/crime-type series.
 - **Crime Count Model**
 - Trained a CatBoostRegressor to forecast daily crime counts for the next two years.
 - Performed hyperparameter tuning to improve accuracy and avoid overfitting, achieving ~80% accuracy.
 - **Correlation**
 - Encoded categorical variables and calculated correlations to find strongly related features.
 - Displayed correlations using a heatmap for clearer interpretation.
 - **Columns Predictions**
 - Based on feature correlations, set a sequential prediction order:
Weapon_Cleaned → Victim_Age_Group → Victim_Gender → Premise_Type.
 - Built a pipeline where each CatBoostClassifier uses both original features and predictions from earlier targets.
 - **Final Output**
 - Generated a CSV containing:
Incident_Date, Area_Name, Crime_Type, Weapon_Cleaned,
Victim_Age_Group, Victim_Gender, Premise_Type.
-

Charts:

- KPIs: Total Predicted Crimes, Lowest-risk Area, Highest-risk Area
- Crime Hotspots symbol map
- Crimes Trendline
- Crime Distribution across areas
- Victim Gender Profile
- Distribution of Age Group

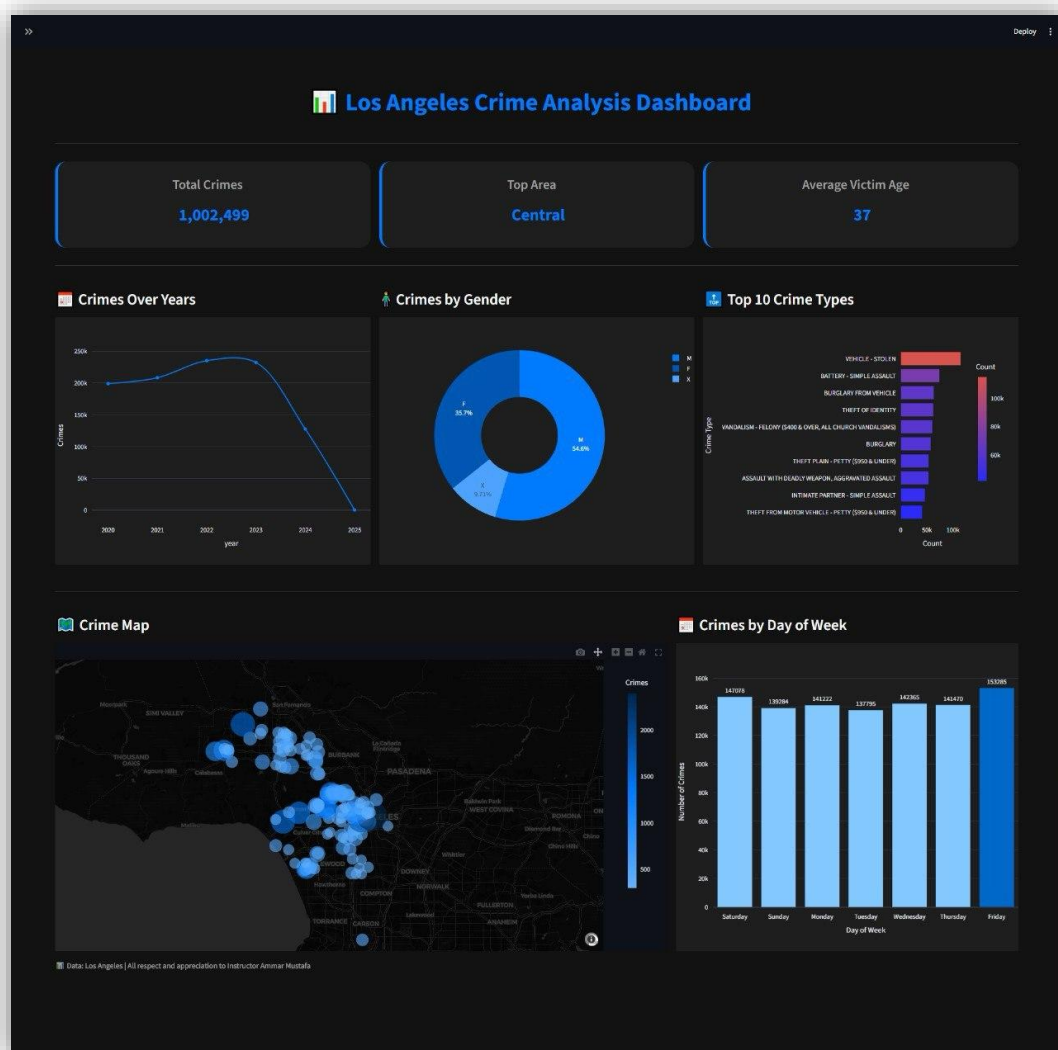
Insights:

- **Total Crimes:** ~309K
- **Highest-risk Area:** 77th Street; **Lowest-risk Area:** Pacific.
- **Hotspots:** Central and southern LA
- **Trend:** Stable with minor spikes.
- **Victims:** Mostly males (55%) and aged 18–30 (~147k).

Link to dashboard:

https://drive.google.com/drive/folders/1ceo8hxEmniOBS5OfqOyhlg2M7wK6CsfS?usp=drive_link

5.5 Python Dashboard (Ziad)



Insights:

- Visualizations using Streamlit
- Crime trends
- Age distribution
- Correlation heatmap
- Weapon relationships

Charts:

- KPI Cards: Total Crimes, Top Area, Avg Victim Age
- Line Chart: Crimes Over Years
- Pie Chart: Victim Gender
- Bar Chart: Top 10 Crime Types
- Bar Chart: Crimes by Day of Week
- Insights: Central records the highest crime concentration, with a clear peak in incidents on Fridays.

Link to dashboard:

[https://drive.google.com/drive/folders/1Ot1nuwCV51PCxif0NOiq6PWefrGkMQ9l?usp=drive link](https://drive.google.com/drive/folders/1Ot1nuwCV51PCxif0NOiq6PWefrGkMQ9l?usp=drive_link)
